

Leona Neftaliem

leonan@stanford.edu | Stanford, CA, USA | [Personal Website](#) | [LinkedIn](#) | [GitHub](#)

Education

Stanford University, Stanford, CA, USA

- Ph.D. Candidate in Environment and Resources Fall 2022 - Present
- Conduct interdisciplinary climate research on urban forests, environmental migration, and air quality, integrating remote sensing, socioeconomic, and environmental data to support environmental policy and strategic investment decisions (Advisors: Drs. Chris Field and Rob Jackson)

George Washington University, Washington, D.C., USA

- B.S. Biology with honors; Minor: Sustainability Fall 2020
- Honors thesis: "How to Get Away with Decomposition: Sunlight driven decomposition of lignin in simulated wood" (Advisor: Dr. Amy Zanne)

Selected Professional Experience

Climate Science Researcher, Suzie Hicks the Climate Chick

Sept. 2024 - Present

- Developed climate science curriculum and audience-tailored educational content for a youth-focused climate media platform with 1.7M+ views; featured on [NPR, April 2025](#) and awarded at the 2025 Climate Film Festival, recognizing excellence in climate-focused media
- Authored grant proposals to secure funding for platform expansion
- Developed curriculum piloted in Los Angeles public schools, with district-wide implementation planned for the 2026 - 2027 academic year

Science Education Contractor, California Academy of Sciences

Feb. 2026 - Apr. 2026

- Designed four science education programs in a compressed 6-week timeline, liaising with researchers, education experts, and senior institutional stakeholders to translate cutting-edge ecological research into accessible content for an institution serving nearly 1 million visitors annually

Biological Science Technician, Smithsonian Environmental Research Center

Jan. 2021 - July 2022

- Managed long-term CO₂ and climate datasets. Developed Arduino-based sensor systems to support field-based environmental monitoring and research

Fellowships and Awards

Stanford Woods Institute's Emerging Environmental Scholar Research Award (\$4,000)

2026

Rising Environmental Leader

2026

E-IPER Research Grant (\$4,000)

2025

Stanford Doerr School of Sustainability Graduate Research Grant (\$1,800)

2025

Stanford Woods Institute Environmental Ventures Project (\$60,000)

2025 - Present

SPIRE Stanford Student Impact Fund Grant (\$6,000)

2025

Smithsonian 'Life on a Sustainable Planet' Research Award (Co-PI; \$75,000)

2025 - Present

Stanford Community Impact Award

2024

Smithsonian 'Life on a Sustainable Planet' Research Award (Co-PI; \$49,188)

2023 - 2024

Knight-Hennessy Scholar (\$306,000)

2023 - Present

National Science Foundation Graduate Research Fellowship (\$138,000)

2022 - Present

Stanford Doerr School of Sustainability Dean's Graduate Scholar (\$100,000)

2022 - Present

Stanford EDGE Fellowship (\$12,800)

2022 - Present

GW Undergraduate Research Fellowship (\$5,000)

2020

GW Sigelman Undergraduate Research Enhancement Award (\$500)

2020

Harlan Undergraduate Summer Fellowship (\$5,000)

2019

Policy and Leadership Experience

Harvard x Smithsonian Center for Astrophysics, BEST AQI Leadership Institute

July 2026

E-IPER 25th Anniversary Planning Committee

May 2026 - Present

Stanford Women's Community Center, STEM Mentor

Oct. 2025 - Present

Knight-Hennessy Scholars, Admission Ambassador

Sept. 2025 - Present

Stanford Sustainability Undergraduate Research Program, Mentor

June 2025 - Aug. 2025

Stanford Doerr School of Sustainability, Recruitment Ambassador

Feb. 2025 - Present

The Building Africa's Cities Summit, Convener

Jan. 2025 - Present

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[United Nations Special Rapporteur](#), Consultant

April 2025

OMG-YA Science Fiction Novel, Researcher

Jan. 2024 - June 2025

Featured in: [Knight-Hennessy Scholars KHeystone Projects, August 2024](#)

Stanford Doerr School of Sustainability Leadership, Peer Wellness Liaison

July 2023 - July 2025

Earthtones Environmental Justice Art Festival, Committee Member

Jan. 2023 - April 2023

R Data Carpentries Workshop, Helper

June 2022

Ethio-Bridge, Mentor

Dec. 2020 - April 2021

Planned Parenthood of Metropolitan, Washington, D.C., Engagement Intern

June 2018 - Aug. 2018

George Washington University Hospital, Volunteer

Jan. 2018 - May 2018

Invited Speaking Engagements

Neftaliem, L. Urban Tree Crown Delineation and the Role of Green Infrastructure in Shaping Migration. Natural Capital Symposium, Natural Capital for Urban Resilience Session, Stanford, CA, June 29, 2026.

Neftaliem, L., Rich, R. L., Brown, D., LaGorga, L., Rosa-Rivera, C., Hedinger, A., Jackson, R. B., Cawood, A. Breathe Baltimore: Community-Engaged Air Quality Monitoring of South Baltimore, Maryland. D.C. Air Research Consortium, Department of Energy and Environment, Remote, February 7, 2025.

Neftaliem, L., Rich, R. L., Brown, D., LaGorga, L., Rosa-Rivera, C., Hedinger, A., Jackson, R. B., Cawood, A. Breathe Baltimore: Community-Engaged Air Quality Monitoring of South Baltimore, Maryland. Baltimore Office of Sustainability, Remote, January 10, 2025.

Rosenfield, M. V., **Neftaliem, L.**, Rich, R. L., Zanne, A. E. The Techno-Ecosphere: Using novel technologies to understand carbon emissions and ecosystem function. Smithsonian Gardens, *Let's Talk Gardens* Webinar, Remote, June 24, 2021.

Rosenfield, M. V., **Neftaliem, L.**, Rich, R. L., Zanne, A. E. Carbon in the Capital: DC Metro carbon dioxide monitoring in the COVID-19 era. Smithsonian Gardens, Remote, October 30, 2020.

Rosenfield, M. V., **Neftaliem, L.**, Rich, R. L., Zanne, A. E. Carbon in the Capital: DC Metro carbon dioxide monitoring in the COVID-19 era. Co-lecture in COVID-19 and the Environment (Walsh School of Foreign Service), Georgetown University, Remote, October 28, 2020.

Posters and Presentations

Neftaliem, L., Rich, R. L., Mady, R., Brown, D., LaGorga, L., Smith, T., Onyango, V., Felix, A., Field, C.B., Jackson, R.B. and Cawood, A., A Community-Engaged Approach to Air Quality Monitoring in South Baltimore. American Geophysical Union, New Orleans., December 2025.

Felix, A. E., **Neftaliem, L.**, Field, C. B. Linking Past Redlining to Present-Day Environmental Inequities in South Baltimore. SURGE Symposium, Stanford, CA, August 2025.

Mady, R., **Neftaliem, L.**, Smith, T., Brown, D., LaGorga, L., Lucchese, V., Quigley, L., Rich, R. L., Cawood, A. Building a community-owned air quality monitoring network in Baltimore, Maryland. Conference for Advancing Participatory Sciences, Portland, OR., May 2025.

Neftaliem, L., Rich, R. L., Brown, D., LaGorga, L., Rosa-Rivera, C., Hedinger, A., Jackson, R. B., Cawood, A. Community-Engaged Air Quality Monitoring of South Baltimore, Maryland. American Geophysical Union, Washington, D.C., December 2024.

LaGorga, L., Smith, T., **Neftaliem, L.**, Noyce, G. L., Rich, R. L. Comparison and Assessment of Low-Cost, DIY Arduino-Based CO₂ Measurement System with Instrument-Measured CO₂ Flux from Automated Chambers over Three Years in a Coastal Wetland. American Geophysical Union, Washington, D.C., December 2024.

Neftaliem, L., Field, C. B., Jackson, R. B. Towards a North American Urban Tree Spatial Dataset: Leveraging Urban Tree Inventories from 30 Cities in North America. American Geophysical Union, San Francisco, CA, December 2023.

Neftaliem, L., Rich, R. L., Noyce, G. L. Can a DIY Arduino-based system accurately measure CO₂ flux from automated chambers? American Geophysical Union, Chicago, IL, December 2022.

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- Neftaliem, L.**, Rich, R. L., Noyce, G. L. Finer Temperature Measurements and GenX Sensors. Global Research Wetland Symposium, Smithsonian Environmental Research Center, Edgewater, MD, March 2021.
- Rosenfield, M. V., **Neftaliem, L.**, Rich, R. L., Zanne, A. E. Carbon in the Capital: DC Metro carbon dioxide monitoring in the COVID-19 era. American Geophysical Union, Remote, December 2020.
- Neftaliem, L.**, Rosenfield, M. V., Zanne, A. E. How to Get Away with Decomposition: Light driven decomposition on lignin in simulated wood. Honors Thesis Seminar, Remote, December 2020.
- Neftaliem, L.**, Rosenfield, M. V., Zanne, A. E. Simulated Wood: Lignin Photodegradation. Harlan Poster Session, Washington, D.C., August 2019.

Publications

- Neftaliem, L., et al. Mapping tree crowns across four North American cities. Manuscript in review.
- Neftaliem, L., et al. Pollution inequity and social vulnerability in Baltimore, Maryland. Manuscript in review.
- Neftaliem, L., et al. Urban Greenspaces, Sense of Place, and Migration Intentions in Climate-Vulnerable U.S. Cities. Manuscript in preparation.
- Neftaliem, L. (2024, October 23). *Breathing life into ghost towns: Harnessing the promise of €1 homes*. Knight-Hennessy Scholar Insights. [Link](#).
- David J. Hayes, Stephen Ferruolo, David Haines, Katelyn McEvoy, **Leona Neftaliem**, Lisa Roberds, Siddharth Sachdeva, Celina Scott-Buechler, Angela Tsao, Katie Vogelheim, Brad Ward, Callie Walker, Benjamin Zehr, Measuring the Carbon (and Other) Benefits of Climate-Smart Forestry Practices (*Policy Lab: Harvesting Climate Benefits from Agriculture and Forestry Practices (808Y)*; *Teaching/Supervising Team: David J. Hayes*). [Link](#).

Skills

Computer: R; Python; Arduino; Bash programming; CRBasic; Google Earth Engine; Jupyter Notebook; ArcGIS Pro; GitHub; EAGLE; LoggerNet; Microsoft Office

Professional: Grant proposal writing, grants management, budget management, stakeholder engagement, cross-sector collaboration, science communication, policy engagement

Languages: English (native), Tigrinya (native)